

1 Fig. 1.1 is a diagram of a molecule of haemoglobin.

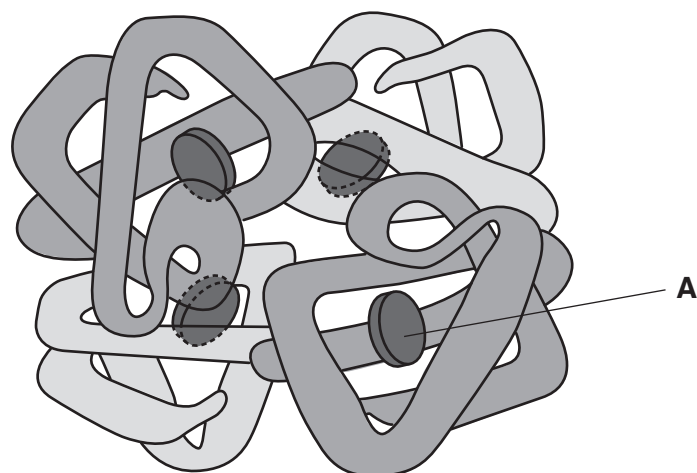


Fig. 1.1

(a) (i) Name the structure labelled **A** on Fig. 1.1.

..... [1]

(ii) State the function of structure **A**.

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..... [1]

(b) Haemoglobin is described as a globular protein. Explain why this protein is described as *globular*.

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..... [2]

- (c) The gene *HBB* codes for the β -globin polypeptide.

State why a polypeptide, such as β -globin, is described as a polymer.

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..... [2]

- (d) A single base change in the DNA of the gene *HBB* results in a change to the amino acid sequence of β -globin. In the sequence, a single glutamic acid is replaced by valine.

Outline the effects of this change in the amino acid sequence of β -globin on the structure and function of a haemoglobin molecule.

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..... [3]

- (e) Haemoglobin interacts with carbon dioxide and carbon monoxide.

Outline the role of haemoglobin in the transport of carbon dioxide.

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..... [3]

[Total: 12]